Demo: Coupled-Coil Showpieces

ChatGPT edition — two safe finales for the course instrument

Visible low-voltage effects without exposed high voltage

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

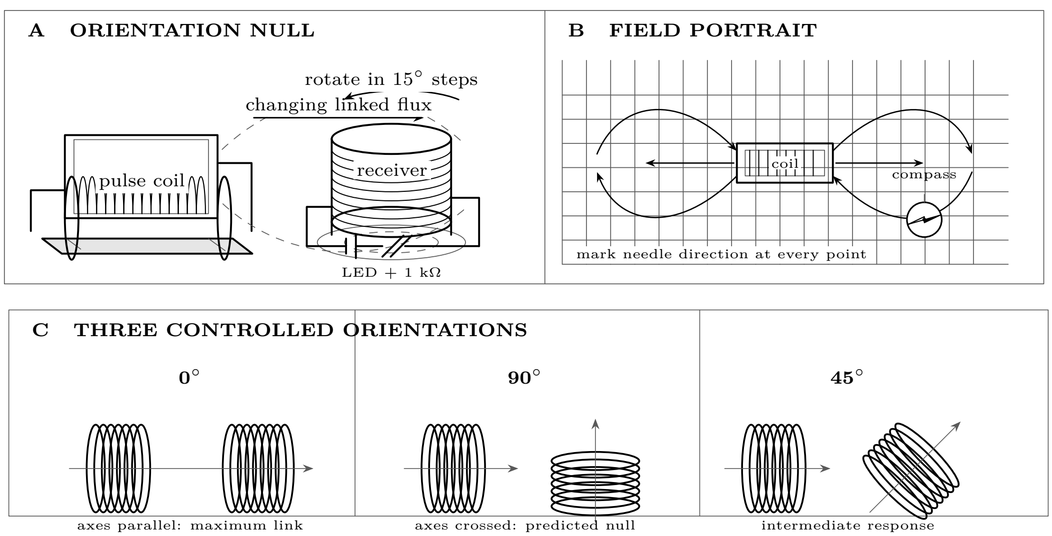
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Completed coupled-coil instrument
- LED with 1 kΩ resistor
- Small compass
- Graph paper

Predict first

At what receiver orientation should the induced indication be weakest?



Investigate

Showpiece 1: find the orientation null

1. Fix the coil centers at one measured separation. Mark the turntable at 0, 15, 30, . . . , 90 degrees. Keep source setting, pulse duration, separation, and lead routing unchanged.
2. At each angle make three identical pulses. Record the meter peak; use the LED only as a visible confirmation. Power off before rotating.
3. Before taking the next row, ask: *Should 45 degrees be nearer the maximum or the null? What observation would prove that our axis label is wrong?*

Angle	Peak 1	Peak 2	Peak 3	Mean	LED / notes
0°					
15°					
30°					
45°					
60°					
75°					
90°					

Graph it. Put receiver angle on the horizontal axis and mean peak on the vertical axis. Add a vertical range bar from the smallest to largest trial. Circle the minimum. Does the curve fall smoothly, or does one point demand a repeat?

Showpiece 2: draw the field portrait

1. Remove the receiver. Tape graph paper beneath the primary and mark its outline, axis, pulse polarity, and a 2 cm grid.
2. At each chosen point place the compass, pulse briefly, and draw the needle direction as an arrow. Repeat once. A disagreement larger than one compass division is a measurement to repeat, not an arrow to average away.
3. Ask midway: *Where are arrows nearly parallel to the coil axis? Where do they curve back? Is the left side the mirror image of the right?*

Grid point	Trial 1 direction	Trial 2 direction	Agree?	Nearby iron / anomaly
(,)				
(,)				
(,)				
(,)				

Join the two results

Place the receiver-axis tracing over the field map at the same center position. At maximum and minimum settings, draw the receiver's area vector. Which setting has the most field crossing its loop area? Write the claim before reading the explanation.

Explain from the evidence

Claim. Receiver output depends on changing magnetic flux through the receiver, not merely on being close to the powered coil.

Evidence. The source pulse, separation, and receiver are unchanged, yet the mean peak changes systematically with angle and reaches a repeatable minimum near crossed axes. The compass portrait shows field direction bending around the primary. Overlaying the receiver shows many field lines crossing its loop area at the strong setting and little net crossing at the null.

Reasoning. The linked flux is $\Phi = \int \mathbf{B} \cdot d\mathbf{A}$, and the induced voltage follows $-N\, d\Phi/dt$. Rotating the receiver changes the dot product without changing the source. Near 90 degrees, contributions through opposite parts of the receiver cancel. Thus geometry changes the measured transfer even when the field is plainly present. The source supplies electrical energy; the changing field transfers a small part to the receiver. Nothing is created at the LED.

Proof standard. The conclusion is earned only if the angle curve is repeatable, its minimum agrees with the area-vector overlay, and the field map has the expected mirror symmetry. An LED is directional and nonlinear, so it supplies immediacy, not quantitative proof.

Change one thing

Symptom	Change one thing, then retest
All angles read zero	Confirm meter range and continuity; reverse receiver leads; shorten separation; verify a source pulse independently.
No clear 90-degree null	Re-center the coils, move leads away from the receiver, remove nearby steel, and confirm the rotation axis passes through its center.
Peaks wander	Standardize pulse duration, allow equal recovery time, secure every lead, and report the full trial spread.
Compass will not settle	Increase distance from steel fasteners and powered leads; compare with the source off; shorten the pulse.
One graph point breaks the trend	Repeat that angle blind, then repeat its two neighbors. Never delete it without a physical reason.

Core comparison. Predict, then test air core versus one shared core. Use the same orientation matrix and add a second curve to the same axes. A valid synthesis states which quantities were controlled, how much the maximum and minimum changed, whether the null angle moved, and how trial spread limits the claim.

Evidence record	
Prediction	______________________
One change	______________________
Observation	______________________
Claim + because	______________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.